

Entanglement as a Resource

Superdense Coding, Quantum Teleportation and QKD

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May 2, 2026

Outline

- 1 Quantum Entanglement
- 2 Superdense Coding
- 3 Quantum Teleportation
- 4 Quantum Key Distribution

Fundamentals

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What is Quantum Entanglement?

Hilbert Space of Combined State

For two quantum systems with Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$, the combined system resides in $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$.

- **Separable Pure State:** A pure state $|\Psi\rangle$ is separable if it can be factored:

$$|\Psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \quad (\text{where } |\psi_1\rangle \in \mathcal{H}_1, |\psi_2\rangle \in \mathcal{H}_2)$$

- **Entangled Pure State:** A state that *cannot* be factored into a simple tensor product.

The Bell States (EPR Pairs)

Four maximally entangled states of two qubits:

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}} (|00\rangle \pm |11\rangle)$$

$$|\Psi^\pm\rangle = \frac{1}{\sqrt{2}} (|01\rangle \pm |10\rangle)$$

The EPR Paradox & Local Realism

- **Einstein, Podolsky, Rosen (1935):** Attempted to show quantum mechanics is an “incomplete theory.”
- **Thought Experiment:**
 - Prepare two particles in an EPR state: $|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$.
 - Separate them by a large distance (Alice and Bob).
 - Alice's measurement instantaneously collapses the combined wavefunction, dictating Bob's state (perfect anti-correlation).
- **Principle of Locality:** No physical influence can travel faster than light c .
- **Spooky Action:** Einstein argued against this non-local instantaneous collapse, advocating for local realism.

Measurement Along Arbitrary Axes

Suppose Alice and Bob measure along an arbitrary axis $\vec{a} = (\sin \theta, 0, \cos \theta)$ in the xz -plane. The spin operator is:

$$\sigma_\theta = \vec{\sigma} \cdot \vec{a} = \sigma_x \sin \theta + \sigma_z \cos \theta = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$$

The eigenvectors of σ_θ are:

$$\begin{aligned} |\theta_+\rangle &= \cos \frac{\theta}{2} |\uparrow\rangle + \sin \frac{\theta}{2} |\downarrow\rangle \\ |\theta_-\rangle &= \cos \frac{\theta}{2} |\downarrow\rangle - \sin \frac{\theta}{2} |\uparrow\rangle \end{aligned}$$

Expressing the singlet state $|\Psi^-\rangle$ in this basis yields the exact same form:

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|\theta_+\rangle \otimes |\theta_-\rangle - |\theta_-\rangle \otimes |\theta_+\rangle)$$

Conclusion: For measurement along any arbitrary shared axis, Alice and Bob will always find perfectly anti-correlated spins.

Mathematical Definition of Hidden Variables

To explain this perfect correlation without non-local collapse, EPR proposed pre-existing properties dictated by local instructions.

Hidden Variables (λ)

Hypothetical, unobservable parameters postulated to supplement the quantum wavefunction $|\psi\rangle$. They assert that the definite outcome of any measurement is governed by a deterministic function $A(a, \lambda)$ depending solely on the local measurement setting a and the hidden state $\lambda \in \Lambda$.

- In this framework, the probabilistic predictions of standard quantum mechanics are simply classical statistical averages over an unknown probability density function $\rho(\lambda)$.
- This reliance on pre-determined outcomes fundamentally relies on **counterfactual reasoning** (asking “what if” for unperformed measurements).

Bell's Inequality

Assume a local hidden variable theory where particles carry pre-determined properties $s_z^A, s_\theta^A, s_\phi^A \in \{\pm 1\}$. Their probabilities are governed by a joint distribution $p(s_z^A, s_\theta^A, s_\phi^A)$.

Because probabilities are non-negative:

$$\begin{aligned} p(++-) + p(-+-) &\leq [p(++-) + p(+++)] + [p(-+-) + p(---)] \\ \implies P(s_\theta^A = +, s_\phi^A = -) &\leq P(s_z^A = +, s_\theta^A = +) + P(s_z^A = -, s_\phi^A = -) \end{aligned}$$

Substituting the known quantum mechanical marginal probabilities for an EPR pair yields **Bell's Inequality**:

$$\sin^2 \frac{\theta - \phi}{2} \leq \cos^2 \frac{\theta}{2} + \cos^2 \frac{\phi}{2}$$

Quantum Violation: Let $\theta = 3\pi/2$ and $\phi = 3\pi/4$.

$$\text{LHS} = \sin^2 \frac{3\pi}{8} - \cos^2 \frac{3\pi}{8} = \frac{1}{\sqrt{2}} \not\leq \text{RHS} = \cos^2 \frac{3\pi}{4} = \frac{1}{2}$$

Derivation of the CHSH Inequality

Consider observables A_1, A_2 (Alice) and B_1, B_2 (Bob) with eigenvalues ± 1 . We define the observable,

$$C = (A_1 + A_2)B_1 + (A_1 - A_2)B_2$$

- **Classical Bound (Hidden Variables):** Because measurements are independent and commutative, either $(a_1 + a_2) = \pm 2$ while $(a_1 - a_2) = 0$, or vice versa. Therefore, $C_{hv} = \pm 2b_i$, yielding:

$$-2 \leq \langle C_{hv} \rangle \leq 2$$

- **Quantum Bound:** Quantum operators do not universally commute. Squaring C yields an operator equation:

$$C^2 = 4\mathbb{I} - [A_1, A_2][B_1, B_2]$$

Since the commutators are bounded ($|\langle [A_i, A_j] \rangle| \leq 2$), we get $0 \leq \langle C^2 \rangle \leq 8$. This yields **Tsirelson's bound**:

$$-2\sqrt{2} \leq \langle C_{qm} \rangle \leq 2\sqrt{2}$$

The CHSH Game

An operational framework demonstrating quantum advantage over local realism.

- **Rules:** A referee sends random bits $x \in \{0, 1\}$ to Alice and $y \in \{0, 1\}$ to Bob. They cannot communicate. They must output bits $a, b \in \{0, 1\}$.
- **Winning Condition:** $a \oplus b = x \cdot y$.

The Classical Limit: A deterministic strategy (e.g., always output $a = 0, b = 0$) satisfies the condition only when $x \cdot y = 0$. Since inputs are random, this bounds the maximum classical win rate:

$$P_{\text{classical}} \leq 0.75$$

The CHSH Game: Quantum Strategy

Alice and Bob share a maximally entangled state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. They map their outputs to spin measurements along specific, misaligned axes:

- **Alice:** If $x = 0$, measures at 0° (z-axis). If $x = 1$, measures at 90° (x-axis).
- **Bob:** If $y = 0$, measures at 45° . If $y = 1$, measures at -45° .

By leveraging the expanded quantum correlation bound ($-2\sqrt{2} \leq \langle C \rangle \leq 2\sqrt{2}$), they calculate joint probabilities that mathematically surpass the classical threshold. The quantum strategy achieves

Tsirelson's bound:

$$P_{\text{quantum}} = \cos^2\left(\frac{\pi}{8}\right) = \frac{1}{2} + \frac{1}{2\sqrt{2}} \approx 0.853$$

Conclusion: The 85.3% win rate physically disproves local hidden variables.

The No-Cloning Theorem

Theorem

There exists no unitary operator U that can take an arbitrary unknown quantum state $|\psi\rangle$ and a blank initialized target state $|e\rangle$, and output two identical copies: $U(|\psi\rangle \otimes |e\rangle) = |\psi\rangle \otimes |\psi\rangle$.

Proof by Contradiction: If U can clone arbitrary states $|\psi\rangle$ and $|\phi\rangle$:

$$U(|\psi\rangle \otimes |e\rangle) = |\psi\rangle \otimes |\psi\rangle$$

$$U(|\phi\rangle \otimes |e\rangle) = |\phi\rangle \otimes |\phi\rangle$$

Taking the inner product of the two equations and using $U^\dagger U = \mathbb{I}$:

$$\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2$$

This requires $x = x^2$, meaning $\langle\psi|\phi\rangle \in \{0, 1\}$. Thus, states must be perfectly identical or perfectly orthogonal.

Implication: Prevents using entanglement to send information faster than light (FTL communication).

Superdense Coding

Rajarshi Giri

Superdense Coding: The Premise

Alice wants to send messages to Bob.

But, she is bored of sending classical bits !!

Question: Can qubit give us any edge over classical bit ? (given that their quantum channel has perfect fidelity)

Holevo Bound

If Alice encodes a classical message into an ensemble of states on an n -qubit system and sends it to Bob, who has no extra side information *correlated with the message*, then Bob can learn no more than n bits of classical information.

No advantage for qubits over classical bits !!

Can we beat Holevo bound ? Just kidding !!

But now, let us change the rules.

Suppose Alice and Bob had bought one member each of an entangled pair from Donald

$$|\Phi^+\rangle_{BA} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Now, The quantum channel is so expensive: Alice can send only **one qubit**.

Yet she wants to communicate **two classical bits**.

Is this possible?

Bennett and Wiesner (1984) showed this is indeed possible !!

Welcome to Dense Coding !!

Encoding (Alice)

Goal: Alice wants to encode two classical bits of information.

Recall she already holds one qubit of the shared state $|\Phi^+\rangle_{BA}$.

She can apply one of four local unitary transformation on her qubit (labelled A):

Gate	Action on basis states	Interpretation
I	$ 0\rangle \rightarrow 0\rangle, 1\rangle \rightarrow 1\rangle$	Do nothing
X	$ 0\rangle \leftrightarrow 1\rangle$	Bit flip
Z	$ 1\rangle \rightarrow - 1\rangle$	Phase flip
XZ	$ 0\rangle \rightarrow 1\rangle, 1\rangle \rightarrow - 0\rangle$	Bit + phase flip

Encoding: Action on the Entangled State

Now What happens to the Global State $|\Phi^+\rangle_{BA}$?

Initial state:

$$|\Phi^+\rangle_{BA} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Example: Apply XZ on Alice (second qubit)

$$(I \otimes XZ)|\Phi^+\rangle_{BA} = \frac{1}{\sqrt{2}} \left[(I \otimes XZ)|00\rangle_{BA} + (I \otimes XZ)|11\rangle_{BA} \right]$$

$$= \frac{1}{\sqrt{2}} \left[|0\rangle_B \otimes XZ|0\rangle_A + |1\rangle_B \otimes XZ|1\rangle_A \right]$$

$$XZ|0\rangle_A = |1\rangle_A, \quad XZ|1\rangle_A = -|0\rangle_A$$

$$= \frac{1}{\sqrt{2}} \left[|01\rangle_{BA} - |10\rangle_{BA} \right] = |\Psi^-\rangle_{BA}$$

Encoding: Summary

Operation (Alice)	Resulting state ($\times \frac{1}{\sqrt{2}}$)
I	$(00\rangle + 11\rangle)_{BA} = \Phi^+\rangle_{BA}$
Z	$(00\rangle - 11\rangle)_{BA} = \Phi^-\rangle_{BA}$
X	$(01\rangle + 10\rangle)_{BA} = \Psi^+\rangle_{BA}$
XZ	$(01\rangle - 10\rangle)_{BA} = \Psi^-\rangle_{BA}$

Global state of the two qubit system $\longrightarrow |\Psi^-\rangle_{BA}$

Now Alice sends her qubit to Bob through a quantum channel (noise free !)

Decoding (Bob)

Bob now holds both qubits B and A .

He can now perform a **joint measurement** on the composite system BA .

Since the four Bell states are mutually orthogonal, Bob can distinguish the current global state $|\Psi^-\rangle_{BA}$ with certainty.

Orthogonality $\Rightarrow$ Perfect Distinguishability

Suppose $\{|\psi_i\rangle\}$ is an orthonormal set. Then one can define

$$M_i = |\psi_i\rangle\langle\psi_i|, \quad \sum_i M_i = I.$$

If the system is prepared in $|\psi_i\rangle$, then

$$p(i) = \langle\psi_i|M_i|\psi_i\rangle = 1,$$

so the states can be distinguished with certainty.

Bell Measurement via Basis Change

In principle, Bob can measure in the Bell basis.

In practice, he measures in the computational basis after a basis change.

- Apply **CNOT** (control: A , target: B) $|b\rangle_B |a\rangle_A \rightarrow |b \oplus a\rangle_B |a\rangle_A$
- Apply **Hadamard** on qubit A $|0\rangle_A \rightarrow \frac{|0\rangle_A + |1\rangle_A}{\sqrt{2}}, \quad |1\rangle_A \rightarrow \frac{|0\rangle_A - |1\rangle_A}{\sqrt{2}}$

Recall our Global state: $|\Psi^-\rangle_{BA}$

$$\begin{aligned} |\Psi^-\rangle_{BA} &= \frac{|01\rangle - |10\rangle}{\sqrt{2}} \\ \xrightarrow{\text{CNOT}_{A \rightarrow B}} & \frac{|11\rangle - |10\rangle}{\sqrt{2}} \\ \xrightarrow{H_A} & |11\rangle \end{aligned}$$

Other Bell states are mapped similarly to $|00\rangle, |01\rangle, |10\rangle$.

Bottom line

Bob identifies the state $\rightarrow$ He backtraces to find which unitary transformation did Alice apply as he was also aware of the mother bell state $|\Phi^+\rangle_{BA} \rightarrow$ Out of four possibilities he identifies 1 outcome $\equiv$ 2 classical bit

THE STORY SO FAR !

- 1 **Preparing and Sharing An EPR Pair:** Prepare a Bell pair

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

And share each qubit to Alice and Bob.

- 2 **Encoding (Alice):** Applies one of $\{I, X, Z, XZ\}$ on her qubit to encode 2 classical bits of message
- 3 **Transmission:** Alice sends her qubit to Bob
- 4 **Decoding (Bob):** Bob makes joint measurement $\Rightarrow$ and retrieves 2 classical bits

Nonlocality of the Message

Before receiving Alice's qubit, Bob has only subsystem B

Start with:

$$\rho_{BA} = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|)$$

We get Bob's density operator by taking partial trace over A :

$$\rho_B = \text{Tr}_A(\rho_{BA}) = \frac{1}{2} \left(\underbrace{\text{Tr}_A(|00\rangle\langle 00|)}_{|0\rangle\langle 0|} + \text{Tr}_A(|00\rangle\langle 11|) + \text{Tr}_A(|11\rangle\langle 00|) + \underbrace{\text{Tr}_A(|11\rangle\langle 11|)}_{|1\rangle\langle 1|} \right)$$

$$\rho_B = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{I}{2}$$

Similarly: $\rho_A = \frac{I}{2}$

Both subsystems are maximally mixed $\Rightarrow$ no local information about the message

The global state is one of four Bell states

Each subsystem alone is maximally mixed:

$$\rho_A = \rho_B = \frac{I}{2}$$

Information is stored in **correlations**, not in individual qubits

Only a joint measurement reveals the message

Say Eve intercepts the transmitted qubit (Alice's):

$$\rho_A = \frac{I}{2}$$

Maximally mixed, reveals no information !!

An adversary must access **both qubits** to extract the message

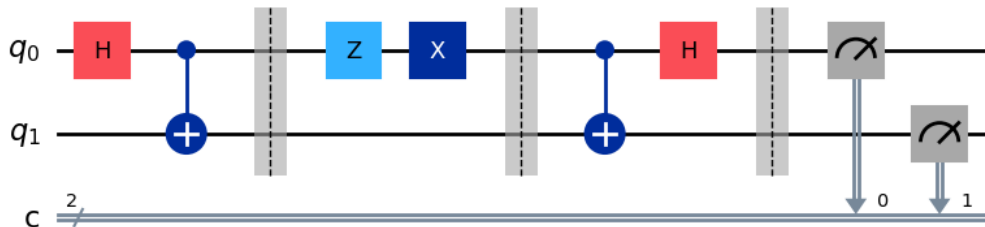
Information is stored in correlations between A and B

Implementation in Qiskit

Superdense coding was implemented in Qiskit.

The circuit was tested on **AerSimulator** and executed on IBM Quantum hardware (*ibm_marrakesh*) using **8192 shots**.

The left part of the circuit explicitly prepares the preshared entangled pair.



Entanglement Preparation (State Evolution)

Recall in the protocol we discussed, Alice and Bob are given a preshared entangled pair. In the circuit, we explicitly prepare that entangled pair using a Hadamard followed by a CNOT

$$|0\rangle_B \otimes |0\rangle_A \xrightarrow{H_A} |0\rangle_B \otimes \frac{|0\rangle_A + |1\rangle_A}{\sqrt{2}} \xrightarrow{\text{CNOT}_{A \rightarrow B}} \frac{|00\rangle_{BA} + |11\rangle_{BA}}{\sqrt{2}}$$

$$|\Phi^+\rangle_{BA} \text{ (entangled pair)}$$

Entanglement Preparation (State Evolution)

Recall: In the protocol, Alice and Bob share a preshared entangled pair. In the circuit, this is explicitly prepared using a Hadamard followed by a CNOT.

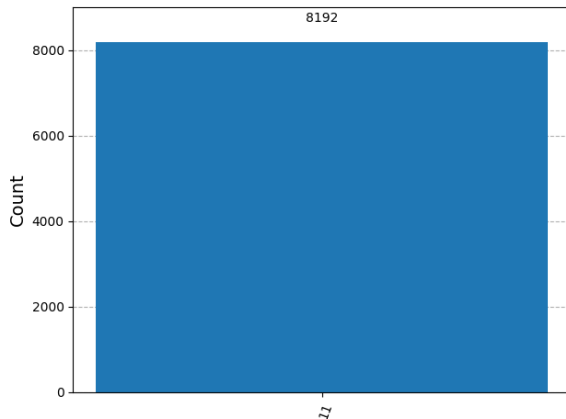
$$|0\rangle_B \otimes |0\rangle_A \xrightarrow{H_A} |0\rangle_B \otimes \frac{|0\rangle_A + |1\rangle_A}{\sqrt{2}} \xrightarrow{\text{CNOT}_{A \rightarrow B}} \frac{|00\rangle_{BA} + |11\rangle_{BA}}{\sqrt{2}}$$

$$|\Phi^+\rangle_{BA} \text{ (entangled pair)}$$

Circuit interpretation: Left $\rightarrow$ Entanglement, Middle $\rightarrow$ Encoding (Alice), Right $\rightarrow$ Decoding (Bob)

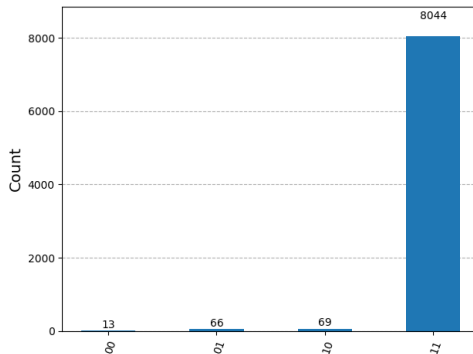
Noise-free simulation

The ideal output is recovered deterministically as **11**.



IBM Quantum Hardware Results

Execution on IBM backend: `ibm_marrakesh` (8192 shots)



Observation

Dominant outcome: **11** appears in **8044/8192** shots.
Small counts in other outcomes arise due to hardware noise.

Quantum Teleportation

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May 2, 2026

Quantum Teleportation: Basic Idea

Suppose Alice has an unknown qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Goal: Transfer this state to Bob using:

- A shared entangled pair (EPR pair)

$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- Classical communication (2 bits)

After the protocol:

- Bob obtains $|\psi\rangle$
- Alice's original state is lost

Outline of the Presentation

In this presentation, we will cover:

1 Quantum Teleportation Protocol

- Theoretical framework
- Step-by-step derivation

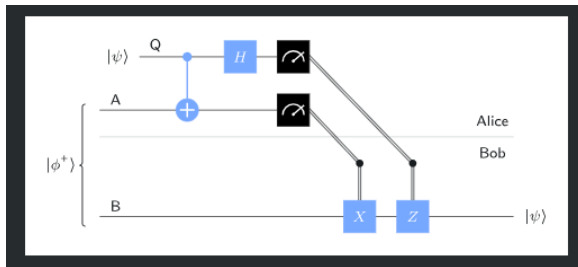
2 Simulation using Qiskit

- Circuit implementation
- Results from simulator

3 Comparison with Real Quantum Hardware

- Execution on IBM Quantum devices
- Noise effects and deviations

Quantum Teleportation Protocol and Circuit



Idea:

- Alice has the unknown qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
- Alice and Bob share an entangled pair $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- Classical bits control corrections on Bob's qubit

Quantum Teleportation Protocol (Step 1)

The total initial three-qubit state is:

$$|\psi_0\rangle = |\psi\rangle \otimes |\beta_{00}\rangle$$

$$\Rightarrow |\psi_0\rangle = (\alpha|0\rangle + \beta|1\rangle) \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

After applying the CNOT gate (control: first qubit, target: second qubit), the state becomes:

$$|\psi_1\rangle = \frac{1}{\sqrt{2}} [\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle]$$

After applying the Hadamard gate on the first qubit, the state becomes:

$$|\psi_2\rangle = \frac{1}{2} \left[\alpha(|000\rangle + |100\rangle) + \alpha(|011\rangle + |111\rangle) \right. \\ \left. + \beta(|010\rangle - |110\rangle) + \beta(|001\rangle - |101\rangle) \right]$$

Grouping of States

Now grouping together the two states of Alice:

$$\begin{aligned} |\psi_2\rangle = \frac{1}{2} & \left[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) \right. \\ & \left. + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle) \right] \end{aligned}$$

If Alice measures her two qubits, then the corresponding output state of Bob becomes:

- The system collapses into one of four possible states
- Each outcome corresponds to a different state of Bob's qubit

Important:

Alice needs to send these two classical bits of information to Bob.

Bob's Correction Operations

Alice's Output	Bob's State	Gate Applied by Bob
00	$\alpha 0\rangle + \beta 1\rangle$	I
01	$\alpha 1\rangle + \beta 0\rangle$	X
10	$\alpha 0\rangle - \beta 1\rangle$	Z
11	$\alpha 1\rangle - \beta 0\rangle$	ZX

Bob applies the corresponding gate to recover the original state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

**The quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
is successfully teleported from Alice to Bob.**

- No physical transfer of the qubit occurs
- Requires entanglement and classical communication

Simulation of Quantum Teleportation in Qiskit

Simulated circuit in Qiskit

The following circuit was implemented and simulated using Qiskit. The unknown state is prepared by applying a unitary operator U on q_0 . Qubits q_1 and q_2 are used to create the entangled Bell pair. After Alice's Bell-state measurement, the classical outcomes control Bob's correction gates X and Z . Finally, Bob applies $U^\dagger$ to verify successful teleportation.

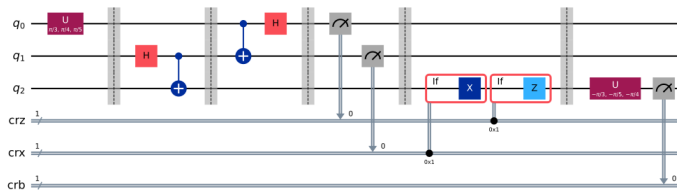


Figure: Qiskit circuit for quantum teleportation and verification.

Quantum Teleportation: Qiskit Simulation Steps

Step 1: State Preparation

- Start with initial state $|0\rangle$
- Apply a unitary operator $U(\theta, \phi, \lambda)$
- This prepares the unknown state:

$$|\psi\rangle = U|0\rangle$$

Step 2: Entanglement Preparation

- Apply Hadamard gate on second qubit
- Apply CNOT to create entangled pair:

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Step 3: Teleportation Protocol

- Apply CNOT and Hadamard on Alice's qubits
- Measure Alice's qubits and send 2 classical bits to Bob

Step 4: Verification

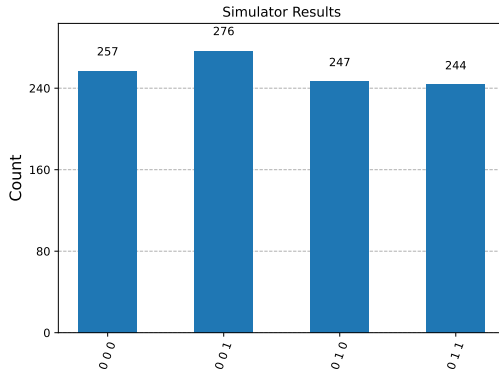
- Bob applies correction gates (X , Z) based on received bits
- Apply U^{-1} on Bob's qubit
- If teleportation is successful : $U^{-1}|\psi\rangle = |0\rangle$

Simulation Results: Histogram Analysis

Results from Qiskit Simulation

The circuit was simulated using Qiskit with a total of 1024 runs (shots). The histogram below shows the measurement outcomes of all qubits.

The leftmost qubit corresponds to Bob's qubit. If the leftmost qubit is measured as 0, it indicates that the teleportation was successful.



Real IBM Quantum Hardware Results

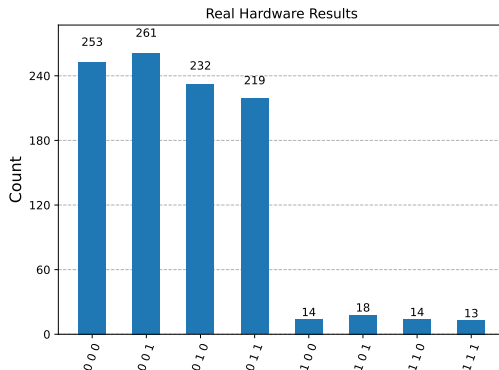


Figure: Measurement outcomes obtained from real IBM quantum hardware for 1024 shots.

Teleportation Success Probability

Conclusion

The leftmost bit corresponds to Bob's verification qubit. If this bit is 0, then successful teleportation has occurred. From the histogram, the successful outcomes are:

000, 001, 010, 011.

$$N_{\text{success}} = 253 + 261 + 232 + 219 = 965.$$

The error outcomes are those where Bob's bit is 1:

100, 101, 110, 111.

$$N_{\text{error}} = 14 + 18 + 14 + 13 = 59.$$

Therefore, the probability of successful teleportation is:

$$P_{\text{success}} = \frac{965}{1024} \approx 0.942 = 94.2\%.$$

The remaining 5.8% error arises from hardware noise, gate imperfections, decoherence, and readout errors.

Why Teleportation Does Not Violate No-Cloning Theorem

No-Cloning Theorem

The no-cloning theorem states that it is impossible to create an identical copy of an arbitrary unknown quantum state:

$$|\psi\rangle \nrightarrow |\psi\rangle \otimes |\psi\rangle$$

What Happens in Teleportation?

In quantum teleportation:

- Alice starts with an unknown state $|\psi\rangle$.
- After performing the Bell-state measurement, her original qubit is **destroyed**.
- Bob's qubit becomes $|\psi\rangle$ only after applying the correct operation.

Key Argument

At no point do Alice and Bob simultaneously possess the state $|\psi\rangle$. The state is **transferred**, not copied.

Teleportation = Transfer of state, NOT duplication

Does the information travel faster than light in quantum teleportation?

- Quantum teleportation involves entanglement and measurement
- The information appears to be transferred instantly
- This raises a fundamental question about causality and locality

Where is the Information?

Before the protocol, the unknown quantum information is in Alice's qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

After the protocol operations (CNOT + Hadamard), the global state becomes:

$$\begin{aligned} |\psi_2\rangle = \frac{1}{2} & \left[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) \right. \\ & \left. + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle) \right] \end{aligned}$$

Key Insight:

- The coefficients α, β no longer belong to a single qubit
- The information is encoded in correlations between all three qubits
- The state has become **non-local**

No Faster-Than-Light Communication

After Alice measures her two qubits, the total state collapses into one of four possibilities. Bob's qubit becomes one of

$$|\psi\rangle, \quad X|\psi\rangle, \quad Z|\psi\rangle, \quad ZX|\psi\rangle.$$

But Bob does not know which one he has. Without Alice's two-bit classical message, Bob's density matrix is still

$$\rho_B = \frac{1}{4} (|\psi\rangle\langle\psi| + X|\psi\rangle\langle\psi|X + Z|\psi\rangle\langle\psi|Z + ZX|\psi\rangle\langle\psi|ZX) = \frac{I}{2}.$$

So Bob still cannot extract any information about $|\psi\rangle$.

Only after Alice sends two classical bits, Bob knows which correction gate to apply. Then he recovers

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

Therefore, teleportation does not send information faster than light.

Quantum Key Distribution

Arkaprava Bose

Basic Idea

- Alice wants to send a message: 1001010101
- She shares a secret key with Bob: 1111111111
- Alice encrypts:

$$\begin{aligned}\text{cipher} &= \text{message} \oplus \text{key} \\ &= 0110101010\end{aligned}$$

- Bob decrypts using the same key : $0110101010 - 1111111111 = \underbrace{1001010101}_{\text{Alice's Message}}$
- Eve (eavesdropper) cannot recover the message without the key

Result

Secure communication is possible if the key is secret.

The Key Distribution Problem

- Reusing the same key is dangerous
- If Eve learns even one message:
 - She can infer the key
 - Future communication becomes insecure
- Solution: Use a new key for every message
- But then:

Main Challenge

How do Alice and Bob generate a shared secret key securely?

This is exactly what QKD solves.

Why Move Beyond Classical Cryptography?

- **Classical Cryptography**

- Security based on **computational hardness**
- Example: factoring (RSA), discrete logarithm
- Assumes limited computational power
- Vulnerable to new algorithms and quantum computers

- **Quantum Advantage**

- Security based on **laws of physics**

Key Principles

No-Cloning Theorem: Quantum states cannot be copied

Measurement Disturbance: Any observation alters the system

Classical: hard to break vs Quantum: fundamentally secure

Classical: hard to break vs Quantum: fundamentally secure

**But can we use it to generate secure keys?
or more specifically**

Key Idea

Can we use entanglement itself to certify security?



E91 Protocol

- Proposed by Ekert (1991)
- Uses **entangled states** for key distribution
- Security based on **Bell inequality**

Key Idea

Eavesdropping disturbs quantum correlations, which can be detected.

Measurement Bases and CHSH Inequality

- The protocol uses the following bases :

$$X, Z, W = \frac{X + Z}{\sqrt{2}}, V = \frac{-X + Z}{\sqrt{2}}$$

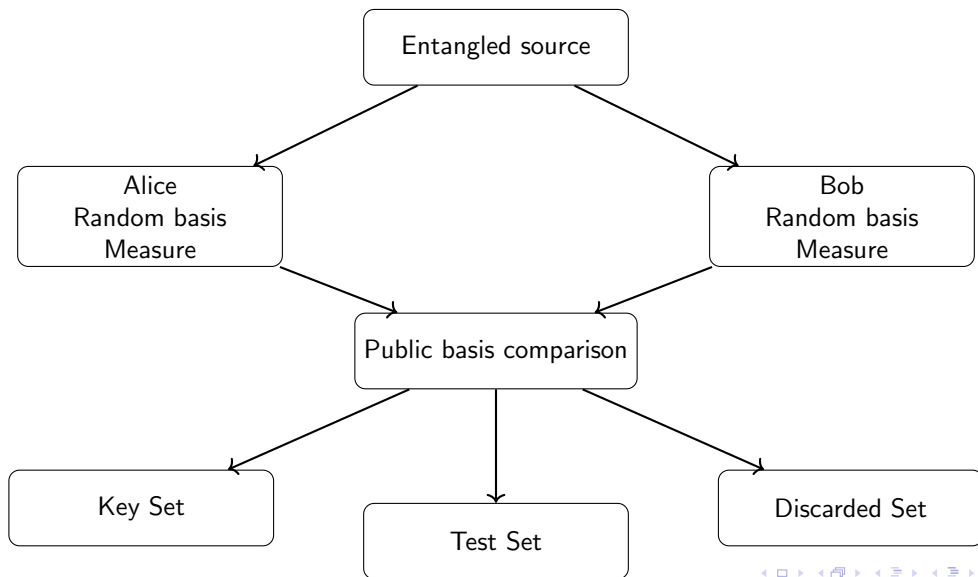
- Alice has access to $\{\mathbf{X}, \mathbf{W}, \mathbf{Z}\}$ and Bob has $\{\mathbf{W}, \mathbf{Z}, \mathbf{V}\}$
- Bell (CHSH) parameter:

$$S = E(a_1, b_1) + E(a_2, b_1) + E(a_1, b_2) - E(a_2, b_2)$$

$$E(a_i, b_j) = \langle a_i * b_j \rangle$$

- Classical bound: $|S| \leq 2$
- Quantum bound: $|S| = 2\sqrt{2}$

E91 Protocol: Steps



3 bases?!

Each of Alice and Bob had 2 bases while we discussed BB84 protocol. Why do we feel the need to change to 3 bases all of a sudden?

- We need to simultaneously generate a key and compute the CHSH parameter $\langle S \rangle$.
- To compute $\langle S \rangle$ we need to assign 2 bases, thus these cannot be used for key generation.
- We use the third key to generate key, if the parameter is well within acceptable range

Advantage

Not all the mismatching bases are discarded unlike BB84 protocol

E91: Partitioning Measurement Outcomes (An Example)

- Alice chooses from: X, W, Z
- Bob chooses: W, Z, V

Key Generation

$$(W_A, W_B)$$

Bell Test (CHSH)

$$(X_A, Z_B), (Z_A, X_B), (X_A, X_B), (Z_A, Z_B)$$

Discarded

$$(X_A, W_B), (Z_A, W_B), (W_A, X_B), (W_A, X_B)$$

E91: Key Generation & Security

- Measurement results split into:
 - **Key set:** matching bases $\rightarrow$ generate key
 - **Test set:** evaluate Bell inequality

- Correlations:

$$E(a_i, b_j) = P_{++} + P_{--} - P_{+-} - P_{-+}$$

- Bell parameter:

$$|S| > 2 \Rightarrow \text{non-local correlations}$$

- If Eve interferes:
 - correlations decrease
 - S drops below threshold

Conclusion

Bell violation certifies secure key generation.

Can Eve Rig the Devices? (Classical)

Since she cannot eavesdrop the encryption, she might try to disrupt the message by rigging the devices!

- Eve may use:
 - preprogrammed outputs
 - hidden variables λ
- Outputs determined as:

$$\text{output} = f(\text{input}, \lambda)$$

- Such classical strategies satisfy:

$$|S| \leq 2$$

Loophole Attempt

Devices could communicate internally to fake correlations

- But E91 Protocol demands:
 - no communication between devices during measurement
 - space-like separation

Quantum Rigging and Monogamy

- Since local strategies fail, she could try something non-local:
 - entangle herself with the system
- Monogamy of entanglement:

$$C_{AB}^2 + C_{AE}^2 \leq C_{A(BE)}^2$$

- If Alice-Bob are strongly correlated:
 - Eve's correlation is limited
 - she cannot gain significant information

Final Insight

Entanglement ensures that Key Distributions become **Device Independent**

Summary

- Classical security relies on computational hardness, Quantum Mechanics enables security to arise from fundamental physics.
- Security in E91 arises from entanglement via Bell non-locality and monogamy and does not rely on trusting the internal workings of measurement devices.

Thank You!